# Supervyse OPF Java Virtual Media User Guide

***Revision 1.0***

***Aug. 2024***

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# 1 Introduction

The user guide describes how to use Virtual Media by the Java client program .

# 2 Prerequisite for executing Java Virtual Media

- Windows 10, 11
  - Install Java 8 or later
    - OpenJDK for Windows is at

      https://learn.microsoft.com/zh-tw/java/openjdk/download

      OpenJDK: Download and install

- Ubuntu
  - Install Java Runtime (JRE > 8)

```
sudo apt install default-jre
```

# 3 Launch the Java Virtual Media Program

Download Java VM Viewer from Virtual Media Web Page:

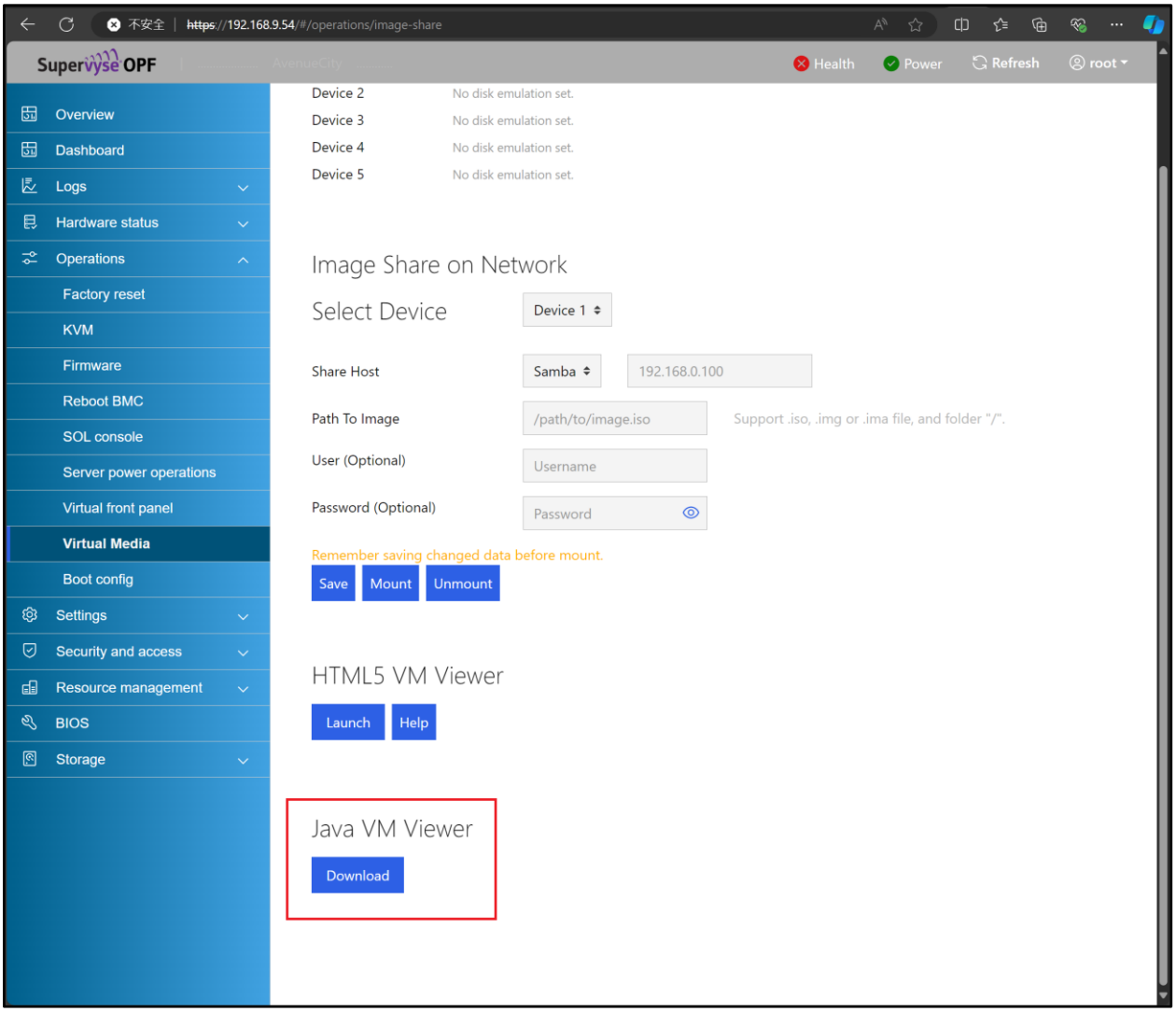


**Figure 3-1 Java VM Viewer download**

Run the downloaded VMViewer.jar by the following command.

```
# java -jar VMViewer.jar
```

The VMViewer UI is shown in the following picture. We can fill in the fields of IP, Port, Username and Password and then press login.

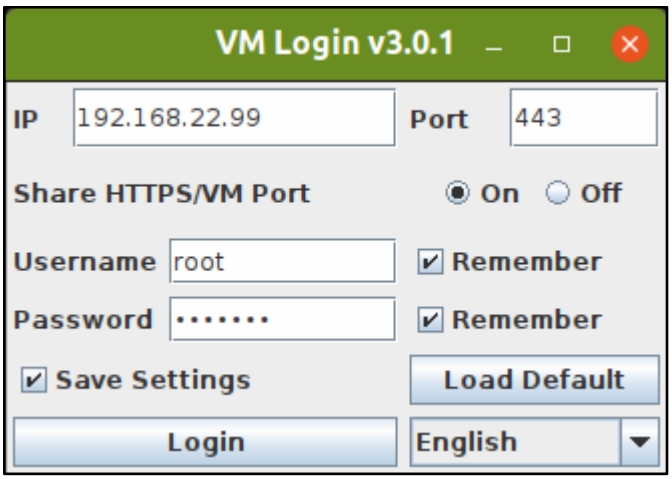


**Figure 3-2 Java VM Viewer login page**

Then, you should see the Virtual Media main program as follows:

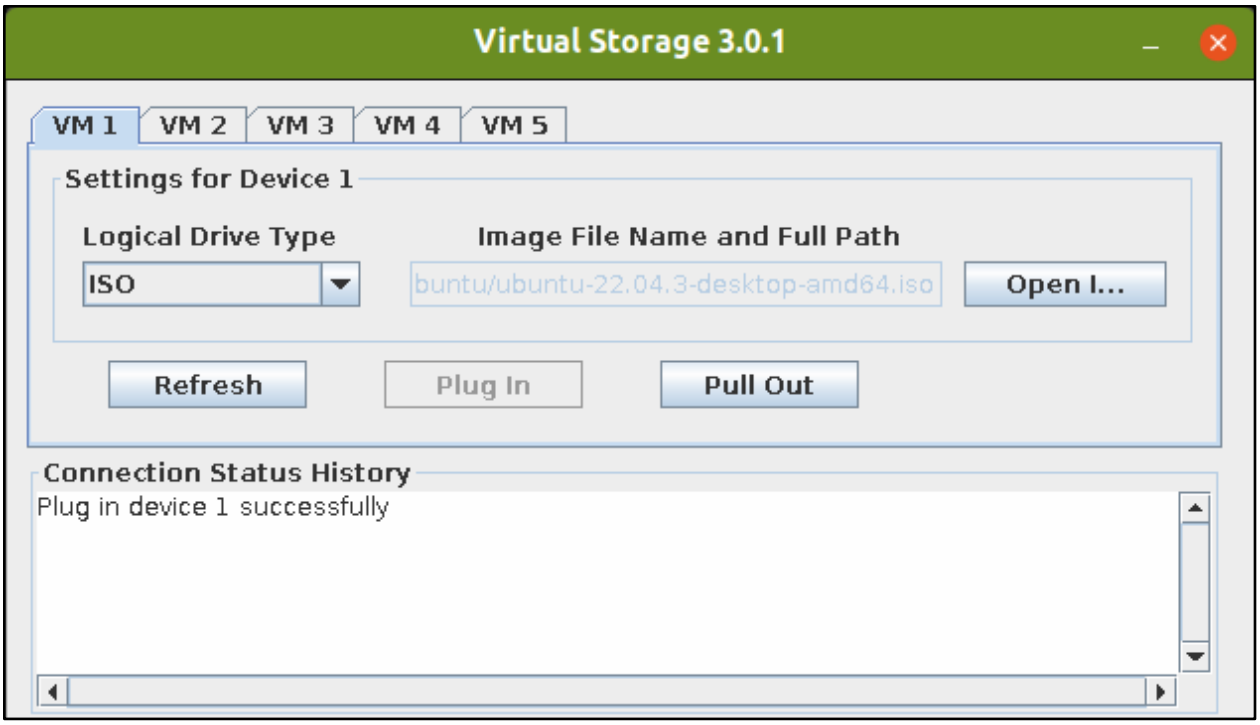


**Figure 3-3 Java VM Viewer main page**

There are a total of six types of Logical Drive Type including IMG/IMA, ISO, Physical CD Drive, Physical USB Drive, SATA/IDE Hard Drive and Folder. All Logical Drive Types will be introduced in the next sections.

***Notice:***

- In order to use USB Drive and SATA/IDE Hard Drive, superuser privilege is required.

```
# sudo java -jar VMViewer.jar
```

- On Aspeed AST2600 and AST2700, you can mount at most 5 virtual devices on the Host at the same time. The maximum number of virtual devices may vary on the capabilities of the BMC chip.
- Click the "VM1" tab to manipulate the selected device, and other tabs respectively.
  The following sections use "VM1" in the examples to demonstrate the virtual media functions.
- The captured images of VMViewer provided in this guide are taken from a system running Ubuntu.

# 4 Features

## 4.1 Mounting a Remote IMG/IMA File

1. Select "IMG/IMA File" from the "Logical Drive Type" drop-down menu.
2. Press the "Open Image" button, then you can browse your file system and select an IMG or IMA image. Then, the path of the file selected will be shown under the text box of "Image File Name and Full Path".
3. Press the "Plug In" button to mount the selected image on the Host. Then, you should be able to see the message "Plug in device 1 successfully" shown in the "Connection Status History" text box if it succeeds.
4. After you finish using the virtual device, press the "Pull Out" button to unmount the image. Afterwards, you should be able to see the message "Pulling out device 1", followed by "Pull out device 1 successfully" shown in the "Connection Status History" text box if it succeeds.

**Figure 4.1-1 Mount a remote img/ima file**

The image is mounted on the host OS like the following picture:

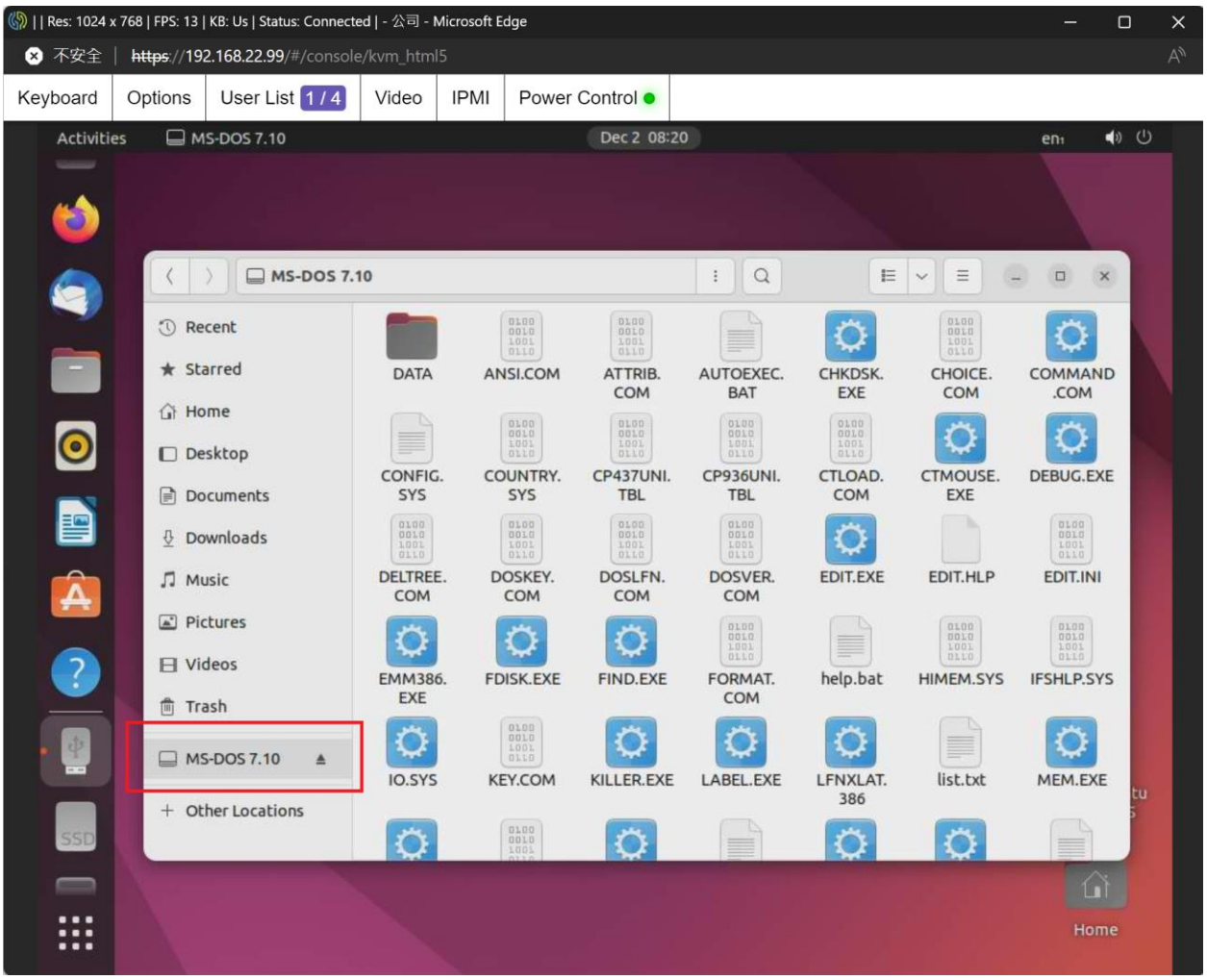


**Figure 4.1-2 Mount img/ima file result**

## 4.2 Mounting a Remote ISO File

1. Select "ISO File" from the "Logical Drive Type" drop-down menu.

2. Press the "Open Image" button, then you can browse your file system and select an ISO image. Then, the path of the file selected will be shown under the text box of "Image File Name and Full Path".

3. Press the "Plug In" button to mount the selected ISO image on the Host. Then, you should be able to see the message "Plug in device 1 successfully" shown in the "Connection Status History" text box if it succeeds.

4. After you finish using the virtual device, press the "Pull Out" button to unmount the image. Then, you should be able to see the message "Pulling out device 1" then "Pull out device 1 successfully" shown in the "Connection Status History" text box if it succeeds.

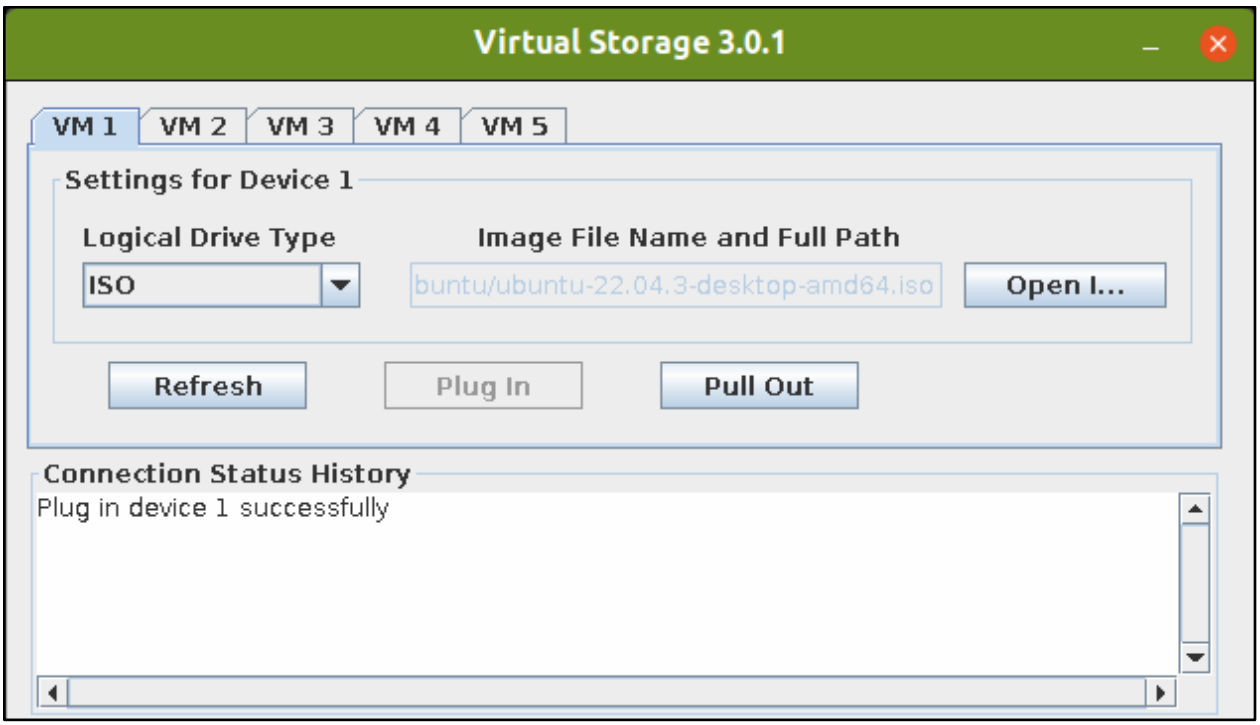


**Figure 4.2-1 Mount remote file**

The iso image is mounted on the host OS like the following picture:

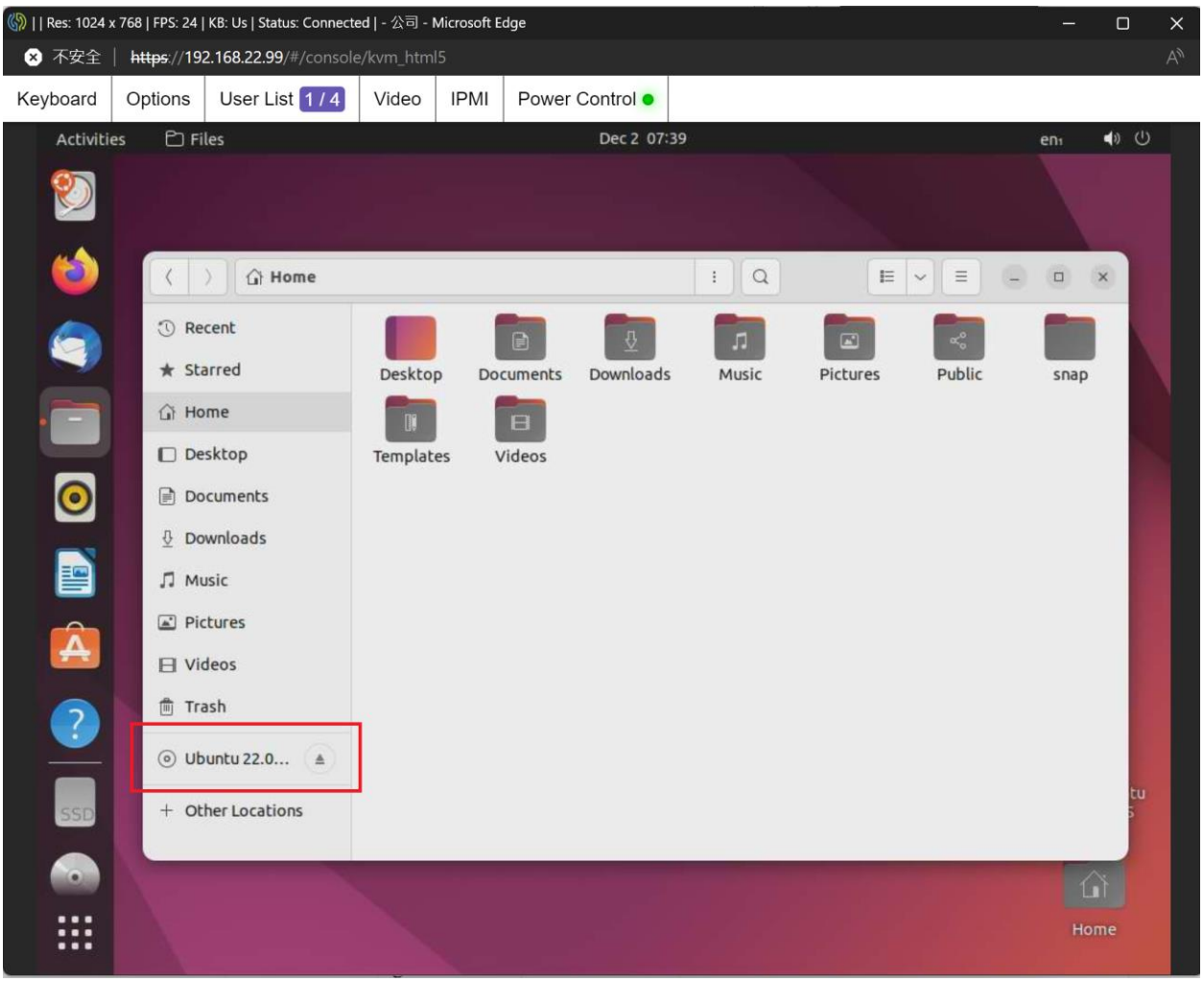


**Figure 4.2-2 Mount remote file result**

## 4.3 Mounting a Remote Physical CD or DVD Drive

1. Select your CD or DVD drive from the "Logical Drive Type" drop-down menu. In this example, our /dev/sr0 is a CD-ROM.

2. Press the "Plug In" button to mount the CD/DVD device on the Host. Then, you should be able to see the message "Plug in device 1 successfully" shown in the "Connection Status History" text box if it succeeds.

3. After you finish using the virtual device, press the "Pull Out" button to unmount the image. Then, you should be able to see the message "Pulling out device 1" followed by "Pull out device 1 successfully" shown in the "Connection Status History" text box if it succeeds.

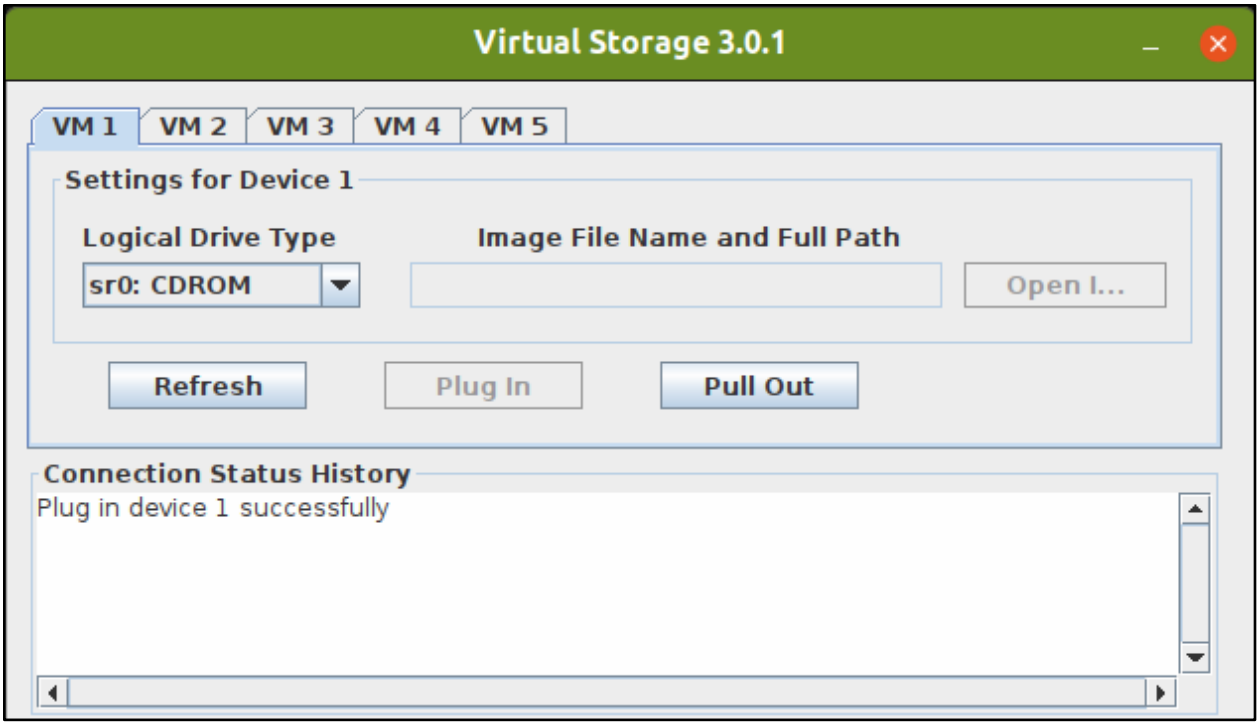


**Figure 4.3-1 Mount CD or DVD**

The CD-ROM is mounted on the host OS like the following picture:

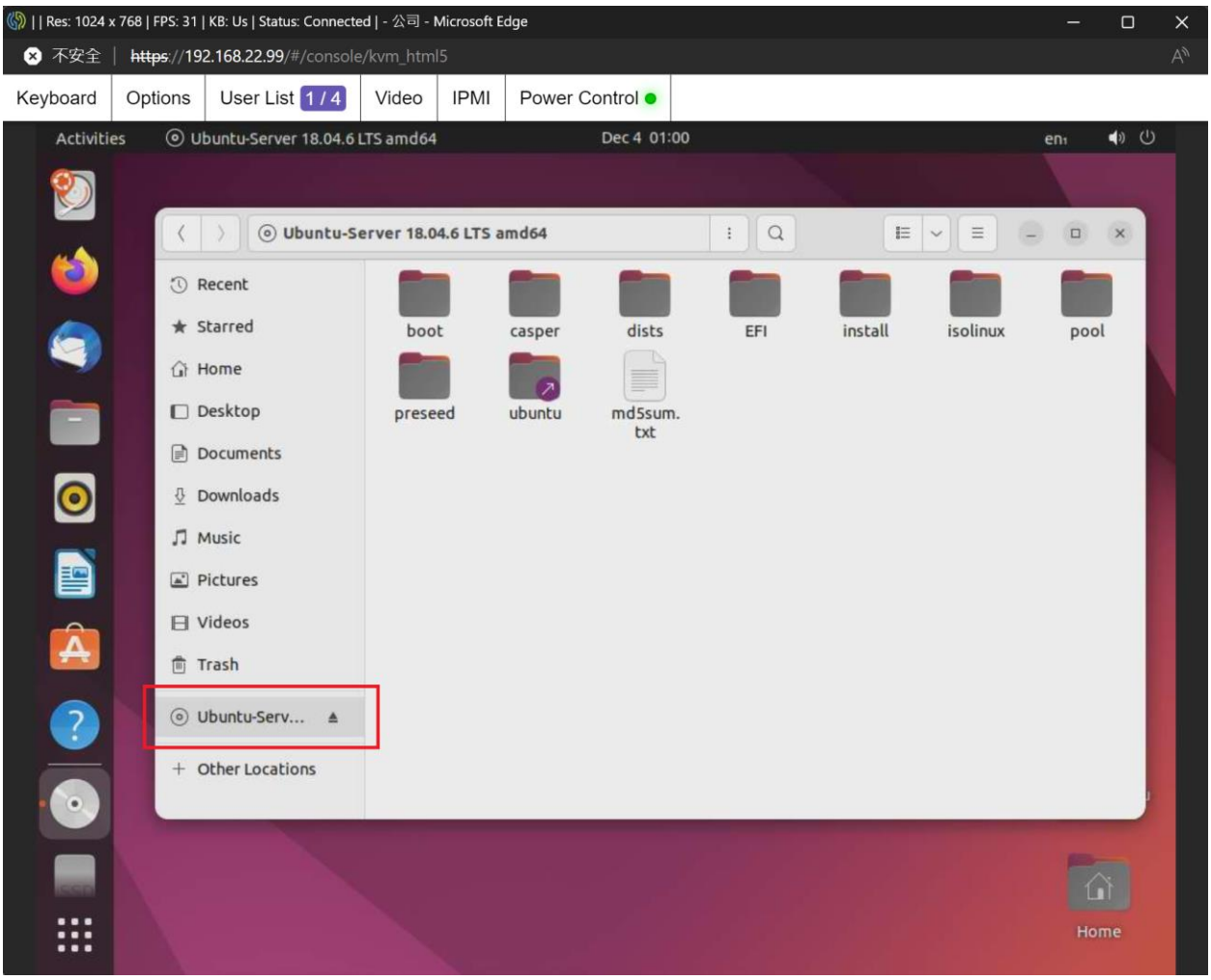


**Figure 4.3-2 Mount CD or DVD result**

## 4.4 Mounting a Remote Physical USB Flash Drive

1. Select your USB Flash drive from the "Logical Drive Type" drop-down menu. In this example, our sdb1 is a USB Flash. If your USB Flash drive is not listed, try pressing the "Refresh" button to allow the program to scan all the USB devices again.

2. Press the "Plug In" button to mount the USB Flash on the Host. Then, you should be able to see the message "Plug in device 1 successfully" shown in the "Connection Status History" text box if it succeeds.

3. After you finish using the virtual device, press the "Pull Out" button to umount the image. Then, you should be able to see the message "Pulling out device 1" then "Pull out device 1 successfully" shown in the "Connection Status History" text box if it succeeds.

**Figure 4.4-1 Mount USB drive**

The USB Flash is mounted on the host OS like the following picture:

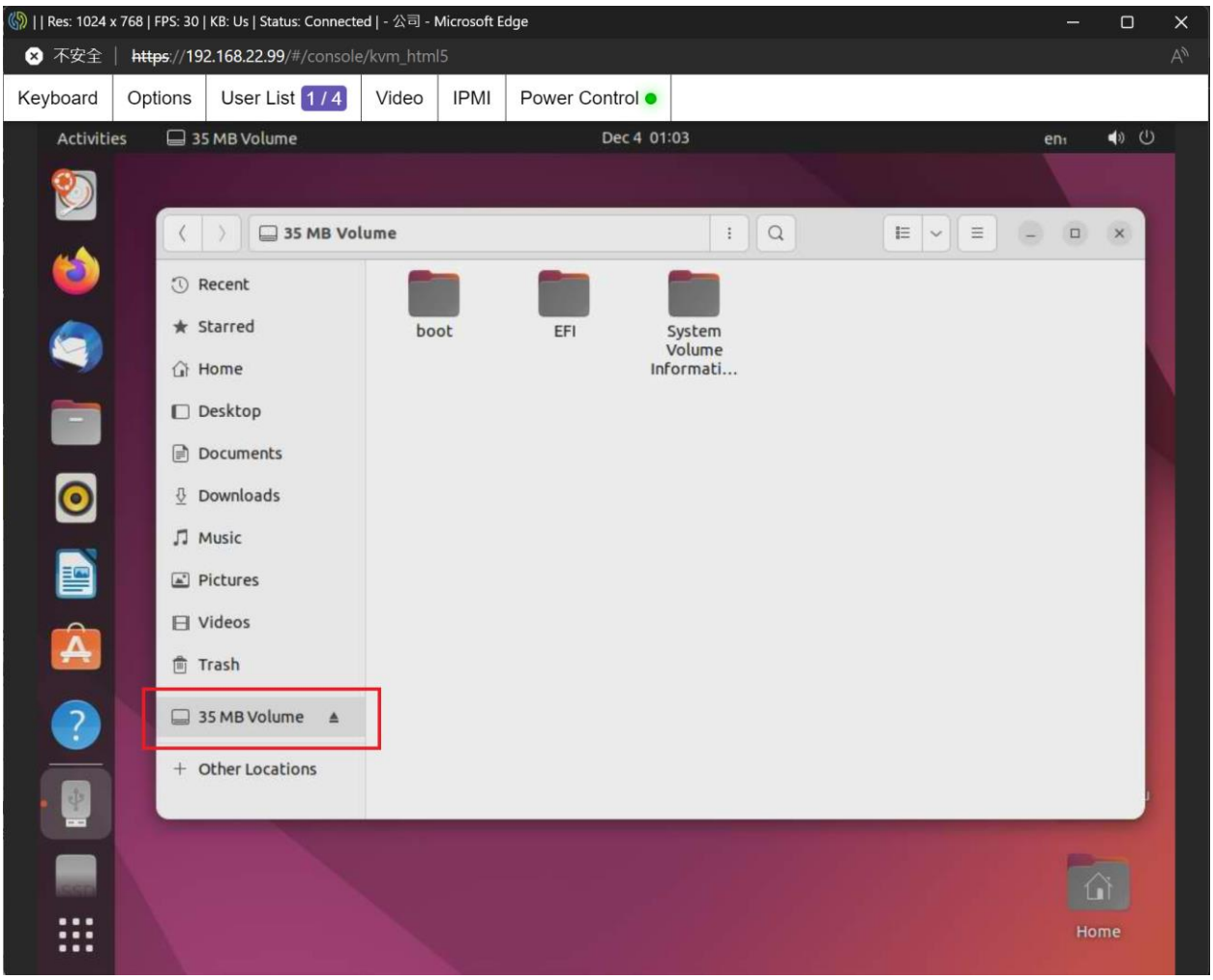


**Figure 4.4-2 Mount USB drive result**

## 4.5 Mounting a SATA/IDE Hard Disk (SATA/IDE HD)

1. Select your SATA/IDE HD drive from the "Logical Drive Type" drop-down menu. In this example, our sda1 is an IDE HD. You might need to click refresh button to see the desired HD.

2. Press the "Plug In" button to mount the SATA HD on the Host. Then, you should be able to see the message "Plug in device 1 successfully" shown in the "Connection Status History" text box if it succeeds.

3. After you finish using the virtual device, press the "Pull Out" button to unmount the image. Then, you should be able to see the message "Pulling out device 1" then "Pull out device 1 successfully" shown in the "Connection Status History" text box if it succeeds.

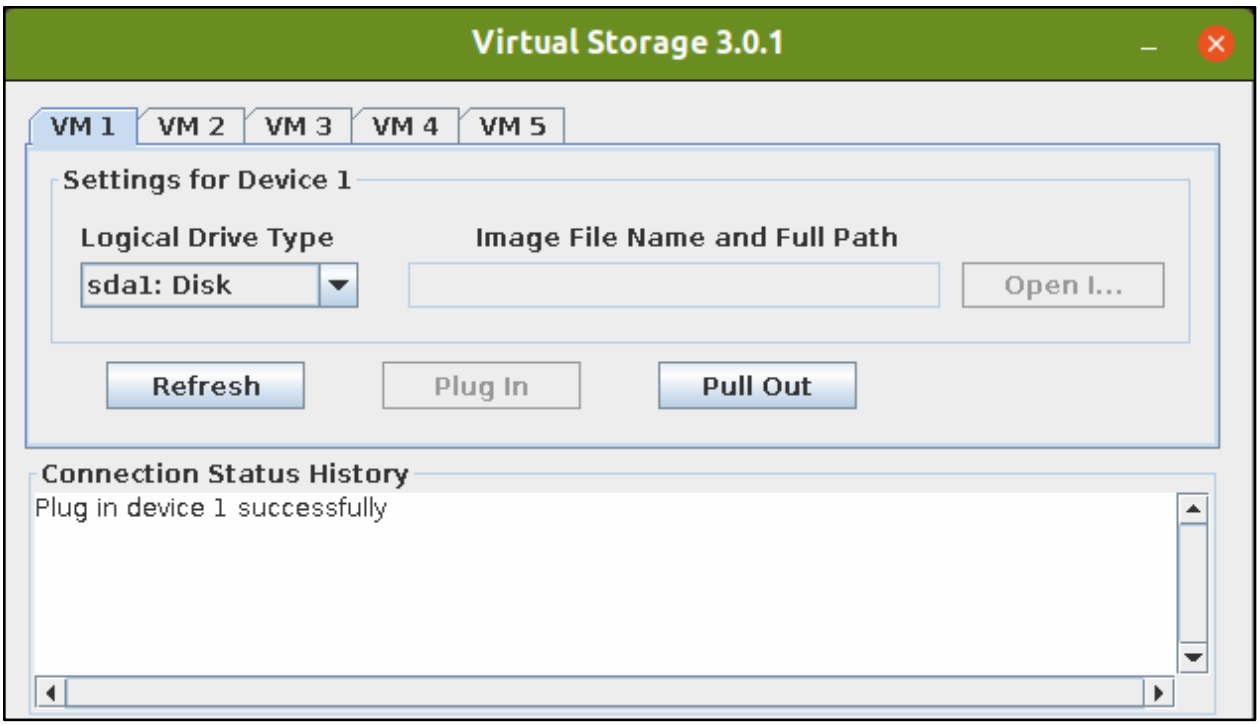


**Figure 4.5-1 Mount hard drive**

The IDE HD is mounted on the host OS like the following picture:

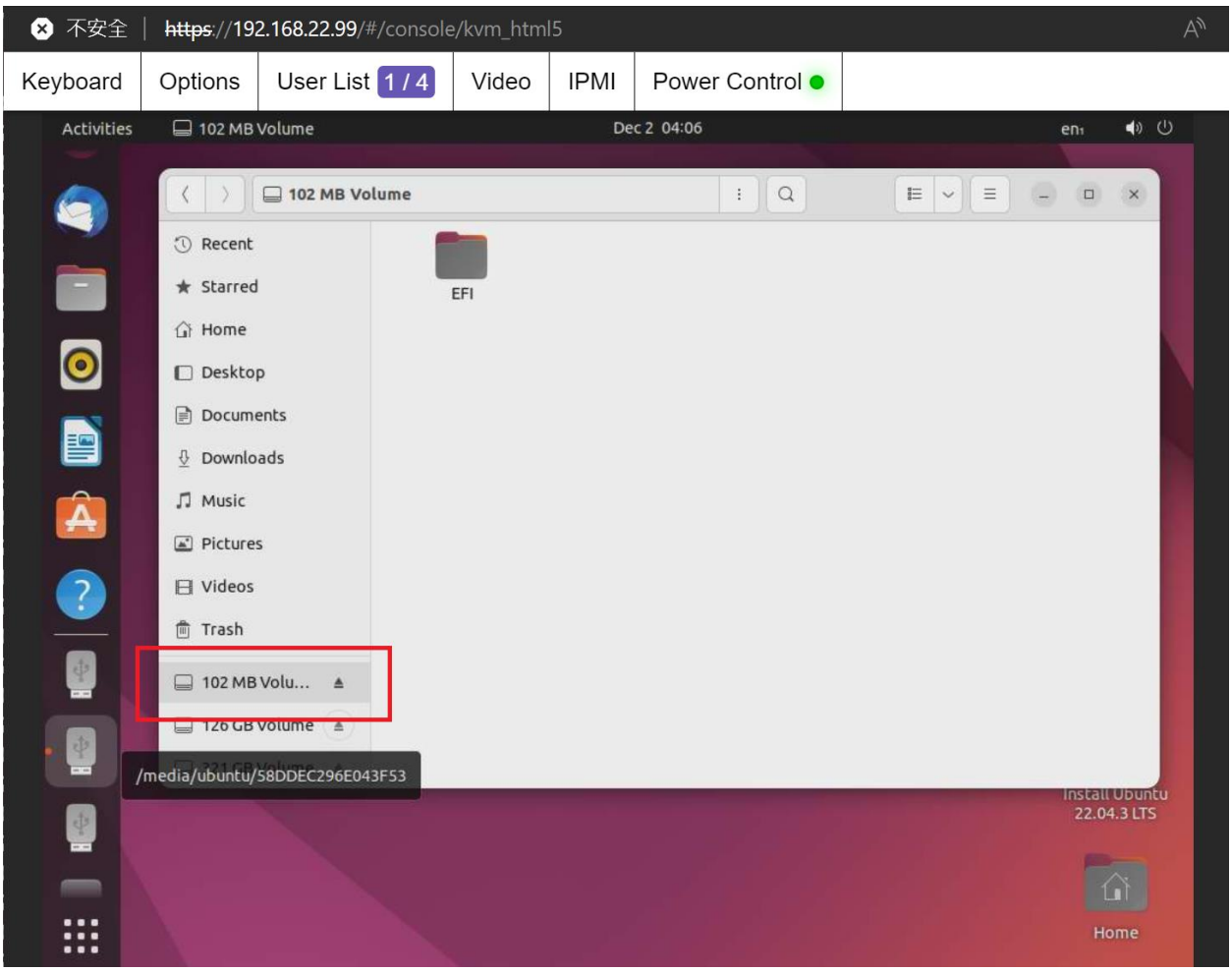


**Figure 4.5-2 Mount hard drive result**

## 4.6 Mounting a Folder

1. Select "Folder" from the "Logical Drive Type" drop-down menu.
2. Press the "Open Image" button, then you can browse your file system and select a folder. Then, the path of the folder selected will be shown under the text box of "Image File Name and Full Path".
3. Press the "Plug In" button to mount the folder on the Host. Then, you should be able to see the message "Device1: VM Plug-In OK!!" shown in the "Connection Status History" text box if it succeeds.
4. After you finish using the virtual device, first unmount the virtual device on host then press the "Plug Out" button to unmount the folder on client. You should see "Device1: VM Plug-Out OK!! Stop!!" shown in the "Connection Status History" text box if the process succeeds.

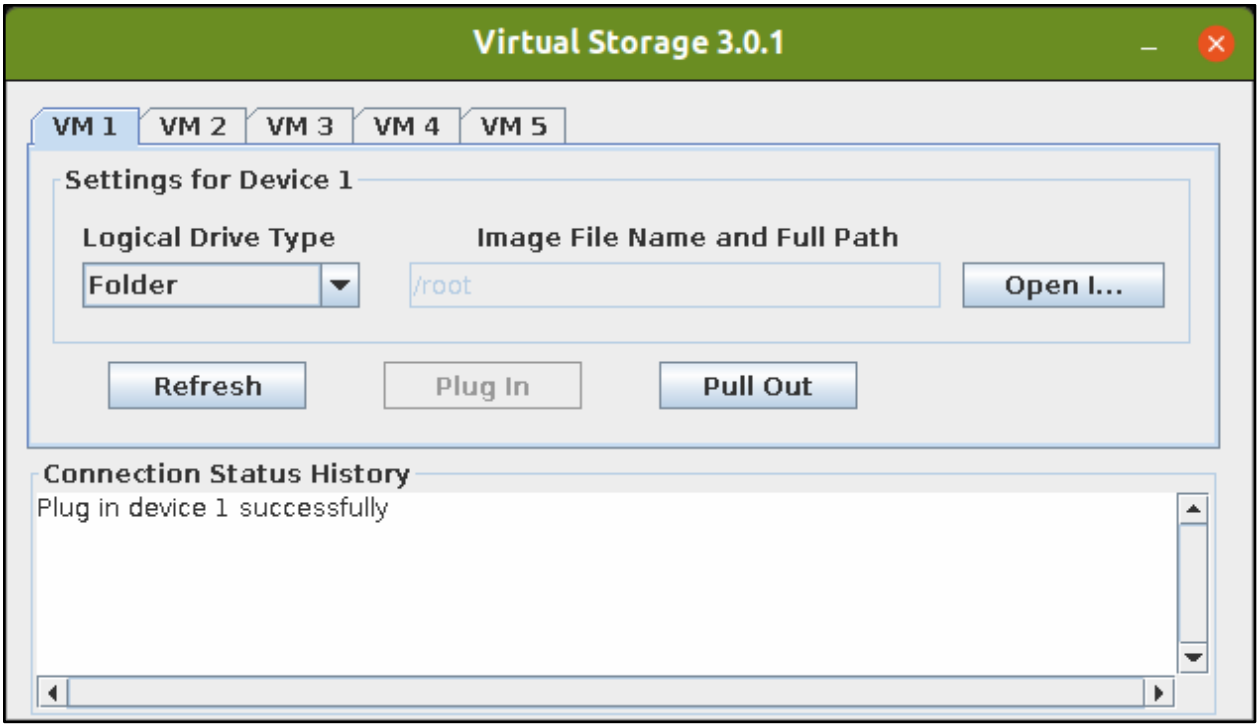


**Figure 4.6-1 Mount folder**

The folder is mounted on the host OS like the following picture:

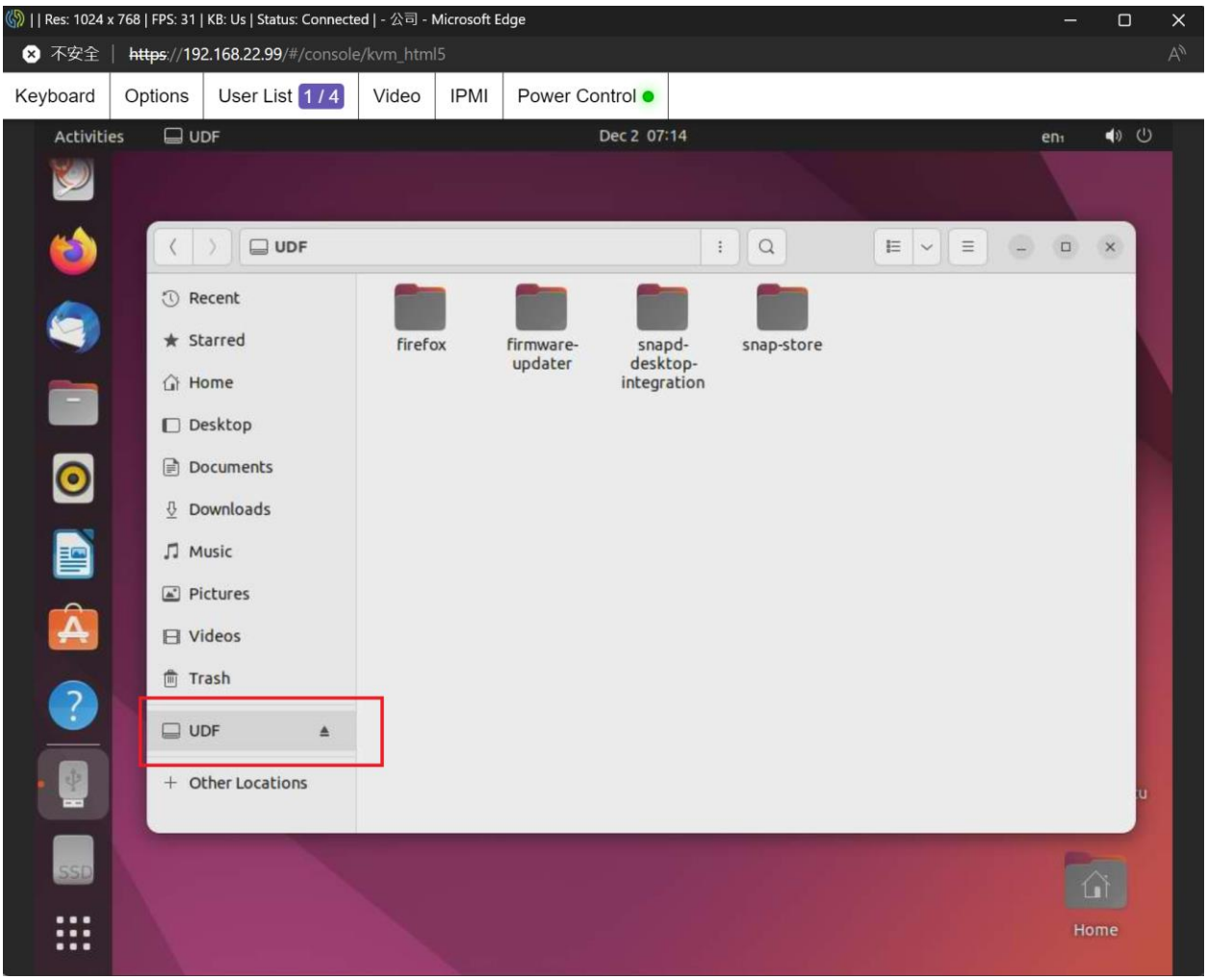


**Figure 4.6-2 Mount folder result**

***Notice:***

- The max total size of files in the folder which you can mount is 1TB, and the max size of a single file is about 40GB.
- The file system of the VM disk is UDF and it is **read-only**.
- Eject the virtual device from the host OS before plugging it out with Java AP to avoid removing the device while the OS is accessing it.
- For security, Virtual Media uses RCM (Remote Control Management) session control. If the web session is terminated, the plug-in function can't be used. But the device which plug-in before web session invalid can still be used.
- The RCM session control will be tested every time a user plugs in a device.

## 4.7 Multi-Language

Select a Language in the dropdown box to change the display language.

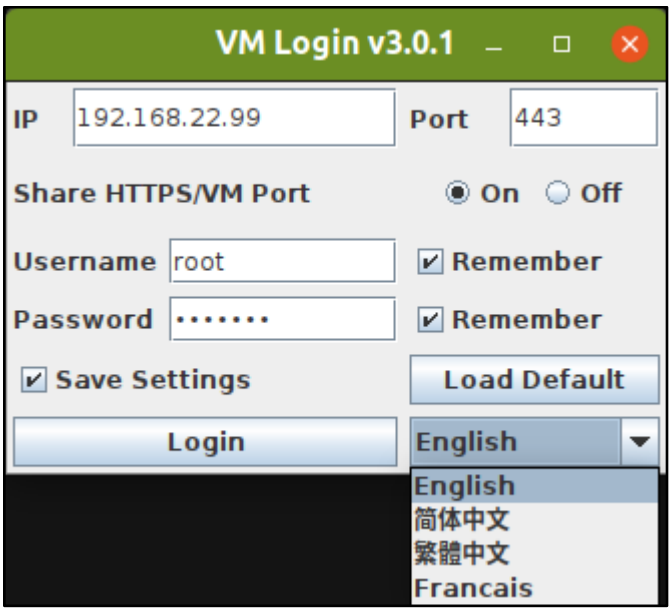


**Figure 4.7-1 Select language**

Then the language of the VMViewer will be changed according to your selection.

**Figure 4.7-2 Select language result**

**Figure 4.7-3 Select language result**

## 4.8 VMCLI Command line tool

Execute the following command with a config file to mount file to host.

```
# java -cp VMViewer.jar tw.com.insyde.vstorage.VirtualStorage test.conf
```

Example of test.conf

```
<?xml version="1.0" encoding="UTF-8"?>
<root>
    <BMC>
        <INDEX>0</INDEX>
        <IP>192.168.0.1</IP>
        <PORT>627</PORT>
        <USERNAME>user</USERNAME>
        <PASSWORD>your password</PASSWORD>
        <FILE_PATH>Path to iso file</FILE_PATH>
    </BMC>
</root>
```

INDEX: The `INDEX` tag specifies the virtual media device to mount. It is a 0-based variable, where an `INDEX` value of `0` corresponds to vm 1 shown on VMViewer application.

IP: IP address of BMC

PORT: Port of BMC

USERNAME: username of BMC

PASSWORD: password of BMC

FILE_PATH: Path of the file to be mounted

When correctly set, the console will print messages like below

```
PS D:\temp> java -cp VMViewer.jar tw.com.insyde.vstorage.VirtualStorage test.conf
Insyde Java VMViewer
Copyright (c) 2013-2024, All Rights Reserved. Insyde Software.
```

```
data path=Folder
data path=ISO
data path=IMG/IMA
NBD Server: file=D:\iso\ubuntu-22.04.3-desktop-amd64.iso
NBD Server: share file=D:\iso\ubuntu-22.04.3-desktop-amd64.iso, port=443
negotiation
transmissionPhase
```

And we can check on Host OS and see if the corresponding VM Device is mounted.

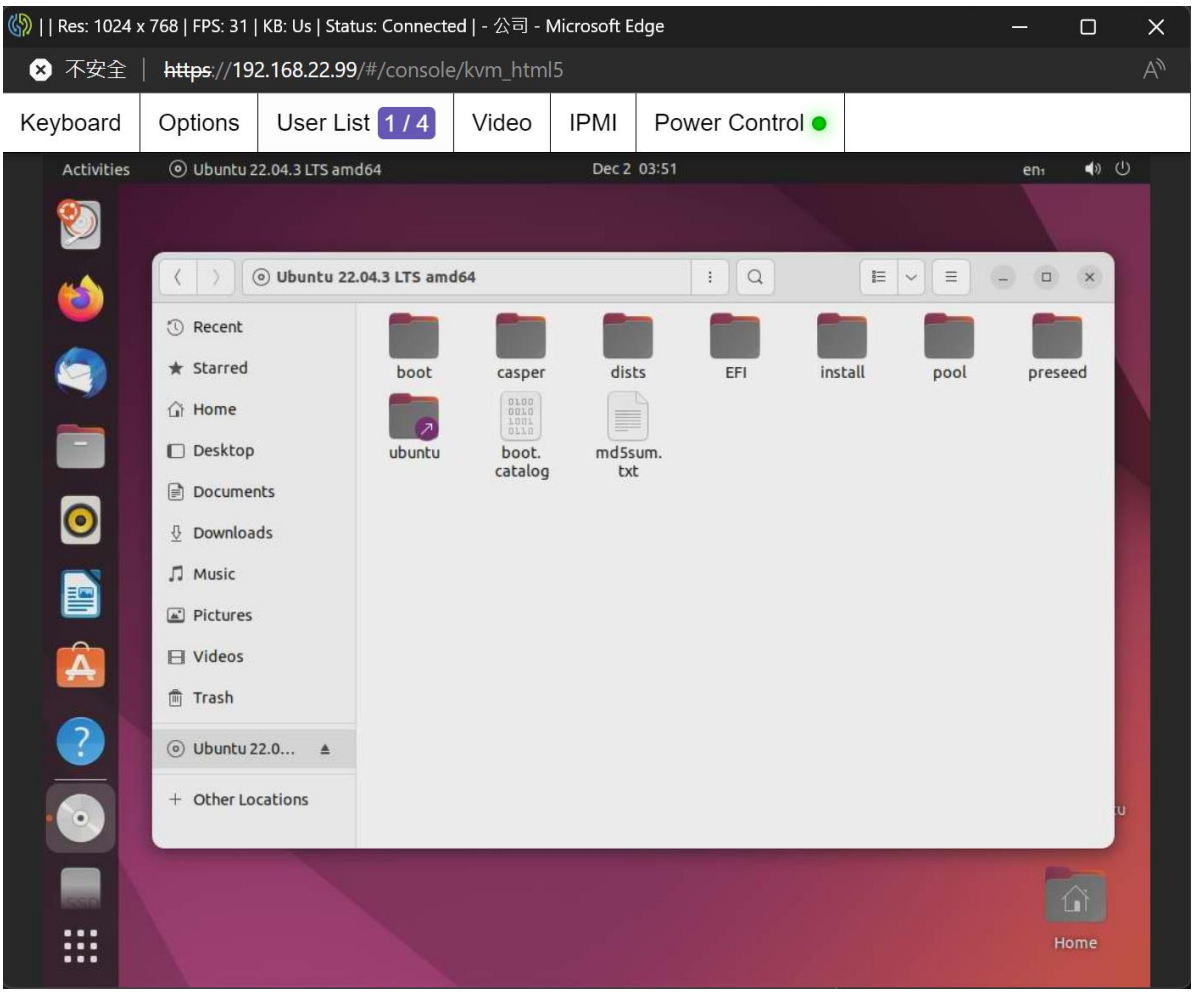


**Figure 4.8 CLI result**

# Appendix

## A. How to Generate Iso Image

Creating an ISO image from a directory containing files. What is genisoimage? genisoimage is a command-line tool used to create ISO 9660 filesystem images, which can be written to CDs, DVDs, or used as virtual disk images. It is especially useful for creating bootable ISOs or archiving directories into an ISO format.

To create a bootable ISO image, you need:

A directory containing the files to be included in the ISO. A boot image file (such as isolinux.bin for Linux distributions).

***Steps:***

1. Install genisoimage

   Ensure genisoimage is installed. If not, you can install it with:

```
sudo apt update
sudo apt install genisoimage
```

2. Obtain isolinux.bin

   You can obtain isolinux.bin from several sources, including the Syslinux project. Here's how

   Visit the Syslinux project page: Syslinux

   Navigate to the Downloads section.

   Download the latest version of Syslinux.

   Extract the downloaded package. You will find isolinux.bin in the isolinux folder within the extracted files.

3. Prepare Your Files

   Organize the files you want in the ISO into a directory. For example: /home/user/iso-dir.

4. Create Bootable ISO

   Run this command to create a bootable ISO:

```
genisoimage -o output.iso -b isolinux.bin -c boot.cat -no-emul-boot -boot-load-size 4 -boot-info-table /home/user/iso-dir
```

   Explanation of Flags:
   -o output.iso: Specifies the name of the ISO file.
   -b isolinux.bin: Specifies the boot image file.
   -c boot.cat: Creates the boot catalog file.
   -no-emul-boot: Ensures no floppy emulation.
   -boot-load-size 4: Sets the number of sectors to load.
   -boot-info-table: Adds a boot info table.
   /home/user/iso-dir: The directory with the files to include.
   This will produce a bootable ISO file named output.iso.

**Verifying the ISO**

After creating the ISO, you can mount it to check its contents:

```
sudo mount -o loop output.iso /mnt
```

The ISO will be mounted at /mnt. You can browse and check its contents.

To unmount the ISO, use:

```
sudo umount /mnt
```

## B. How to Generate IMG Image

To create an IMG file from a physical device (such as a USB stick or hard disk), use the dd command as shown below and obtain the img file:

```
sudo dd if=/dev/sdX of=/path/to/output.img bs=4M status=progress
```

Explanation of Flags:

if=/dev/sdX: The input source is your device, such as `/dev/sdb` representing a USB stick or hard disk.
of=/path/to/output.img: The output file is `output.img`.
bs=4M: Specifies a block size of 4MB for read/write operations.
status=progress: Shows progress information.

If you want to create an IMG file from a directory, first create a blank image, then mount the file system onto this image.

***Steps:***

1. Create an empty IMG file

```
dd if=/dev/zero of=/path/to/output.img bs=1M count=1024
```

Explanation of Flags:

if=/dev/zero: Uses `/dev/zero` to generate a stream of zero bytes, creating an empty file.
of=/path/to/output.img: Specifies the path and name of the output IMG file.
bs=1M: Sets the block size to 1 megabyte.
count=1024: Determines the size of the IMG file (1024 blocks * 1MB = 1GB).

2. Format the IMG file with the ext4 file system

```
mkfs.ext4 /path/to/output.img
```

3. Create a directory to mount the IMG file:

```
mkdir /mnt/tmp
```

4. Mount the IMG file to the temporary directory using a loop device:

```
sudo mount -o loop /path/to/output.img /mnt/tmp
```

5. Copy the desired files from your source directory to the mounted IMG file:

```
sudo cp -r /path/to/source/* /mnt/tmp
```

6. After copying the files, unmount the IMG file to save the changes

```
sudo umount /mnt/tmp
```